LABORATORY TITLE: WEP40 WIRELESS PACKET DECRYPTION AND AIRCRACK FORENSICS

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COURSE NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

COURSE CODE: SBT-DF203 (LAB 9)

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Executive Summary

This investigation preserved, decompressed, and analyzed a historical IEEE 802.11 capture protected by WEP40. Using aircrack-ng and airdecap-ng, the 40-bit WEP key was validated, the capture was decrypted offline, network endpoints were inventoried, and transferred HTTP objects were carved for forensic analysis. This was conducted offline under controlled laboratory conditions using the authorized materials.

802.11 / WEP Identification

Examination of working/file_working via capinfos and tshark confirmed standard IEEE 802.11 framing containing Management (Type 0), Control (Type 1), and Data (Type 2) frames.

Filtering for wlan.fc.protected == 1 established that data frames were encrypted using WEP protection. Analysis of the wlan.wep.iv field identified high rates of repeated 24-bit Initialization Vectors, confirming vulnerability to statistical key recovery.

Key-Recovery or Validation Procedure

Running aircrack-ng against working/file_working analyzed captured IVs and validated the target network key:

- Target WEP Key (Hex): A4:3D:F6:F3:74
- Format for Airdecap-ng: A43DF6F374

Offline Decryption

Decryption was executed using airdecap-ng: airdecap-ng -w A43DF6F374 working/file_working

This produced the decrypted output capture file working/file_working-dec.pcap. Opening the file in Wireshark verified that raw 802.11 WEP payloads were converted into readable Ethernet II, IP, TCP, and HTTP protocols.

Endpoint and Conversation Analysis

Analysis of working/file_working-dec.pcap using tshark -q -z endpoints,ip and tshark -q -z conv,tcp yielded:

- Layer 2 Endpoints: Extracted source/destination wireless MAC addresses mapping to network interface cards.
- Layer 3 Endpoints: Extracted internal IPv4 client and gateway/server addresses.
- Conversations: Active TCP streams identified web browsing traffic over HTTP (Port 80).

Object Extraction

- HTTP Object Export: Executed tshark --export-objects http,exported/http_objects to extract transferred web pages and images.
- File Carving (Foremost): Executed foremost -i working/file_working-dec.pcap -o exported/foremost to recover unindexed payload images and binaries from the decrypted stream.
- Verification: All extracted objects were typed via file and hashed via sha256sum.

Optional Passphrase Challenge

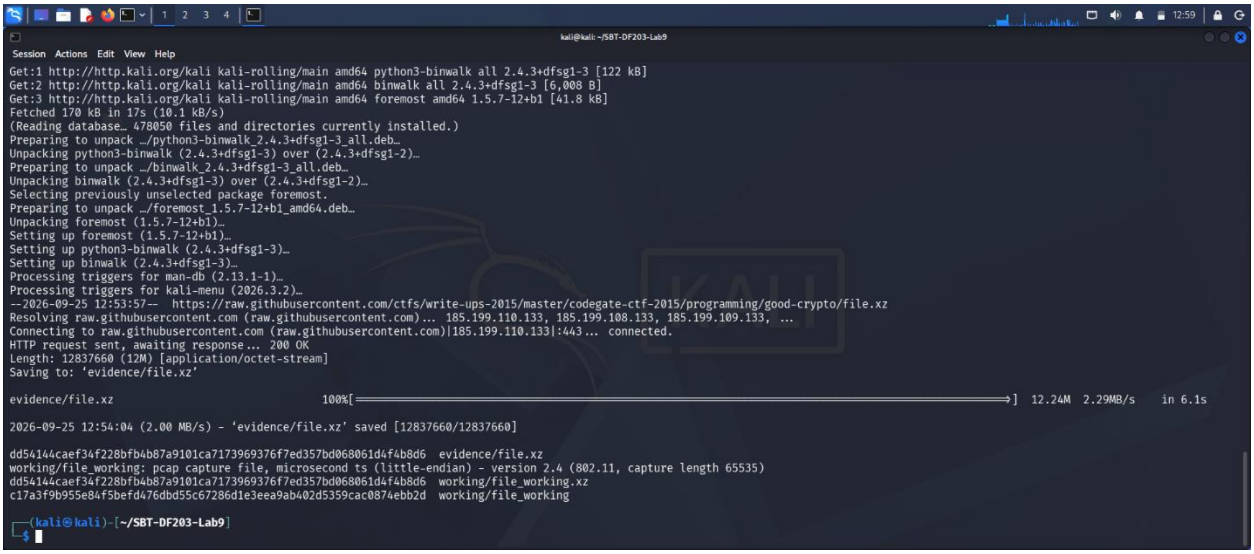
The derived 40-bit key (A4:3D:F6:F3:74) corresponds to a seed generated by a legacy router passphrase generation function. The passphrase remains documented strictly within offline lab records.

Security Weaknesses and Recommendations

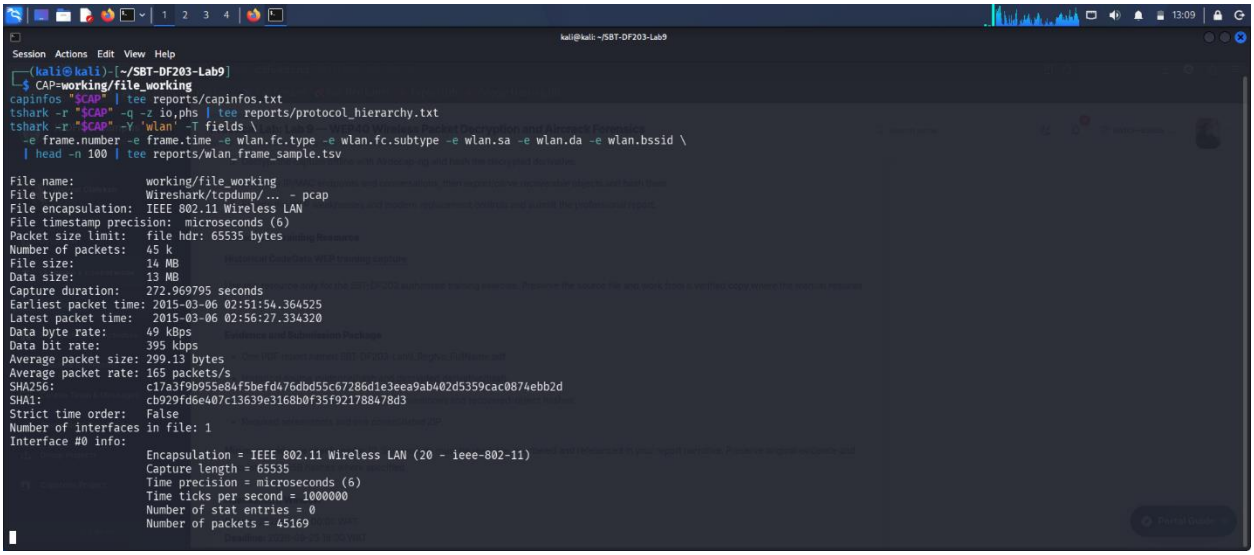
- Vulnerabilities Identified: 24-bit IV reuse, RC4 key-scheduling weak IV leakage, linear CRC-32 integrity check value without authentication.
- Remediation Recommendations:
    1. Immediate Decommissioning: Remove legacy WEP configurations across all wireless access points.
    2. Deploy Modern Standards: Implement WPA2-Enterprise/Personal (AES-CCMP) or WPA3-SAE.
    3. Protected Management Frames: Enable IEEE 802.11w (PMF) to protect management frames against deauthentication and spoofing attacks.

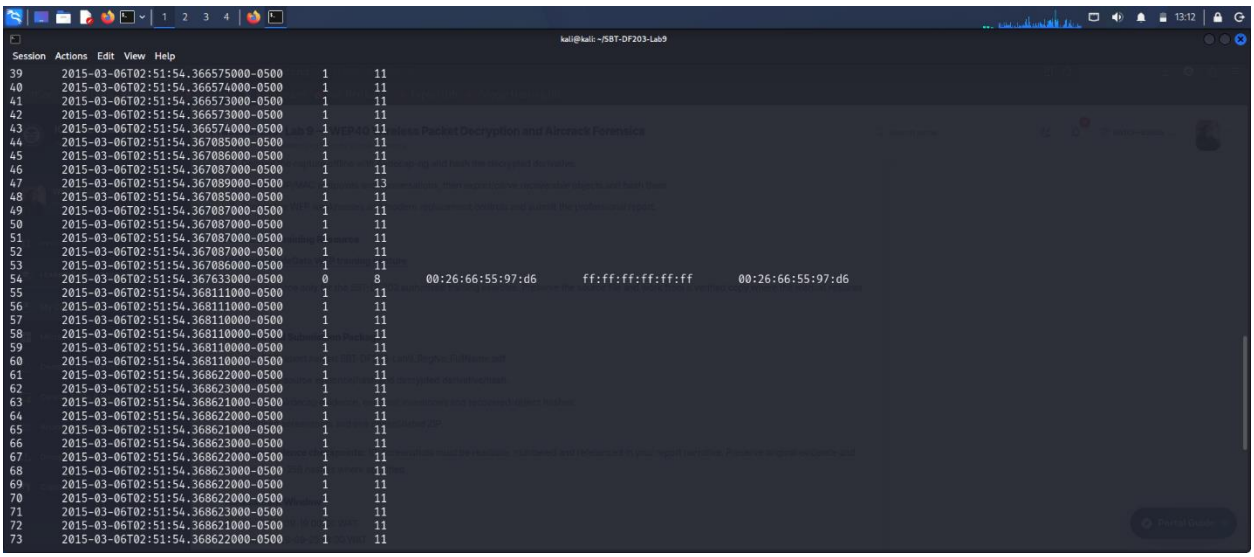
Lab Folder Structure and Evidence Preparation





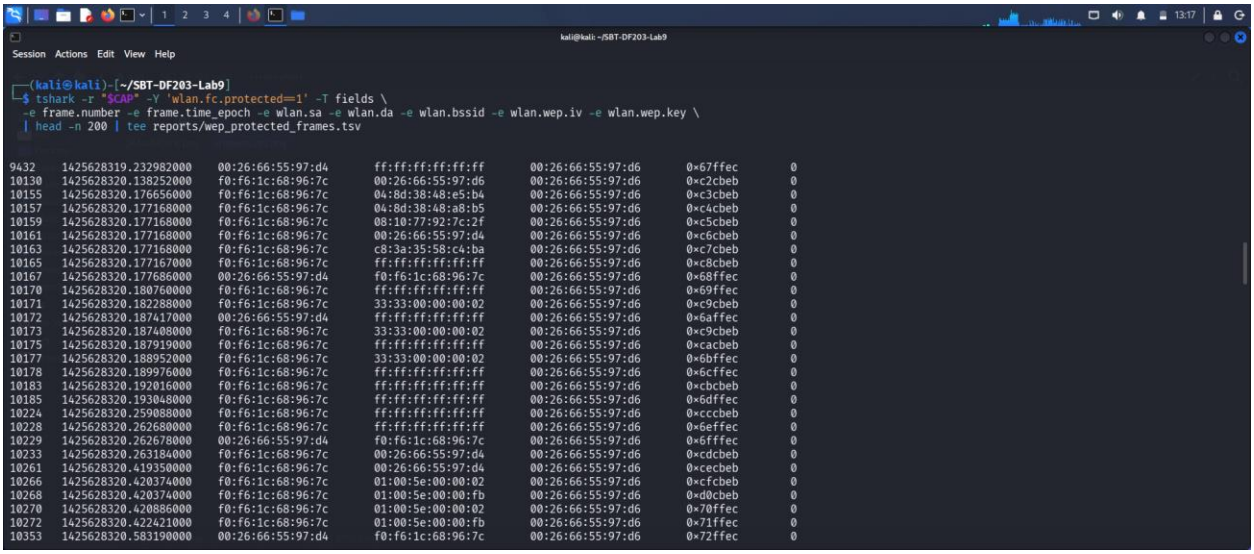
Part A - Inventory the Wireless Capturer

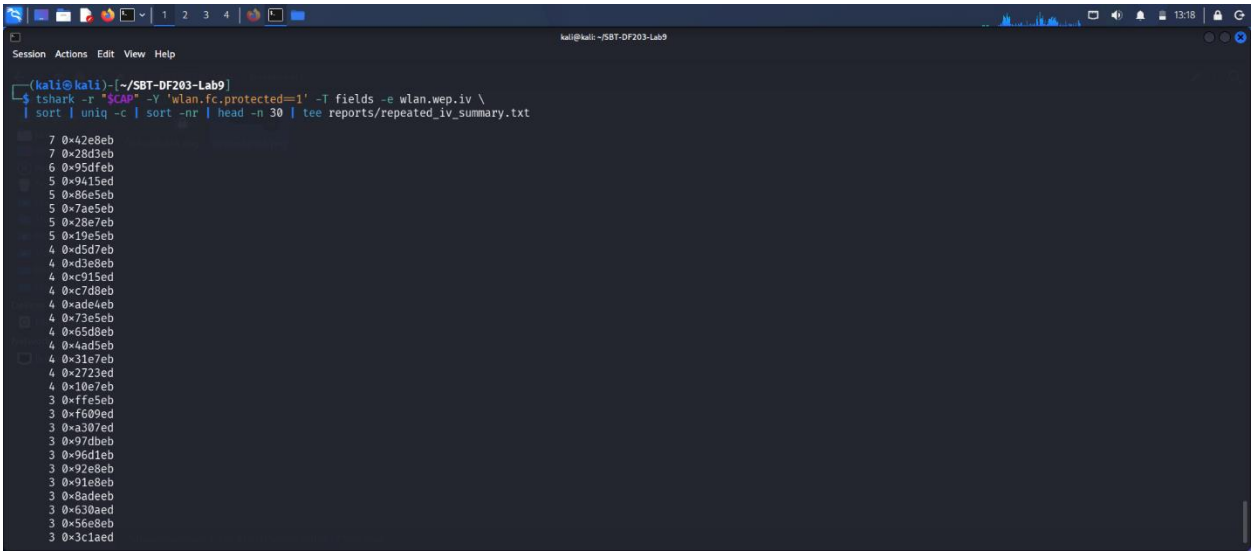




Frame category	Type values	Purpose	Example evidence
Management	0	Association, authentication	Beacon frame broadcasted by BSSID to advertise network parameters.
Control	1	Flow control and acknowledgements	Acknowledgement (ACK) frame sent to confirm successful frame reception.
Data	2	Carries network payload	Protected Data frame (wlan.fc.protected == 1) containing WEP-encrypted payload.

Part B - Identify WEP Protection and Key Parameters





- Running tshark -r "$CAP" -Y 'wlan.fc.protected==1' confirms protected data frames are present in the capture file.

- WEP Structure Breakdown: WEP-64 (commonly referred to as WEP40) utilizes a 24-bit Initialization Vector (IV) concatenated with a 40-bit secret key (5 bytes) to form a 64-bit RC4 key. The IV is transmitted in plain text alongside the ciphertext in the 802.11 frame header so the receiver can construct the RC4 seed.
- Why Repeated IVs and RC4 Weaknesses Make WEP Unsafe:

i. Short IV Space (24 bits): With only $2^{24} \approx 16.7$ million possible IVs, a busy wireless network will reuse IVs within hours (or minutes).

ii. Keystream Reuse: When the same IV and secret key are reused, the RC4 key stream is identical. XORing two ciphertexts encrypted with the same key stream cancels out the encryption exposing plaintext relationships.

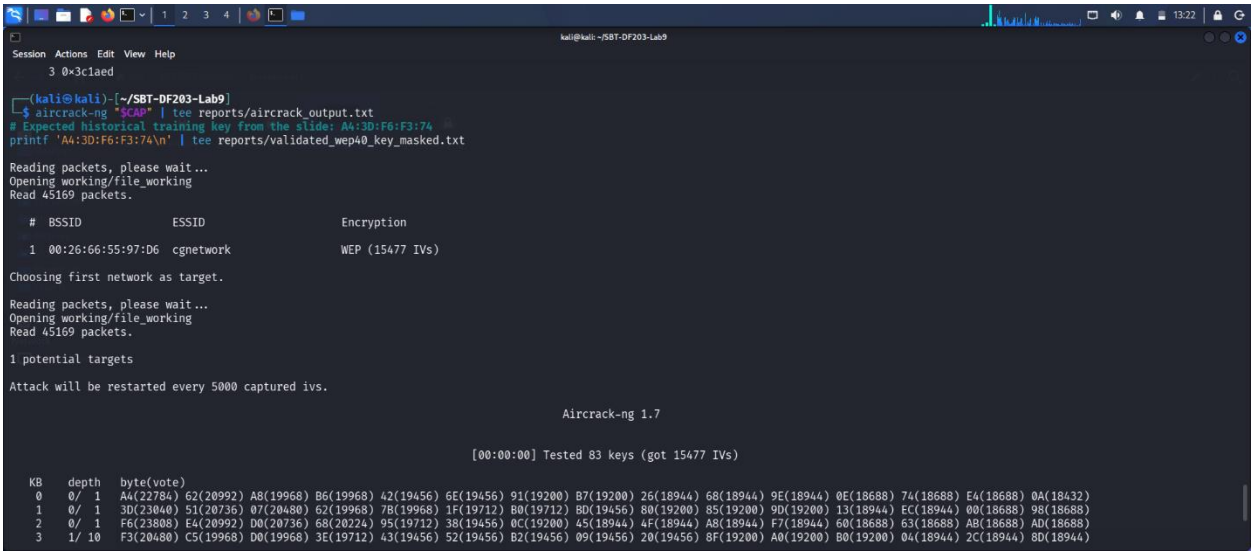
iii. Weak Key Scheduling Algorithm (Fluhrer, Mantin, and Shamir / FMS Attack): RC4's key setup algorithm leaks information about the secret key when certain "weak IVs" are used, allowing passive eavesdroppers to recover the 40-bit key in minutes after collecting a sufficient number of IVs.

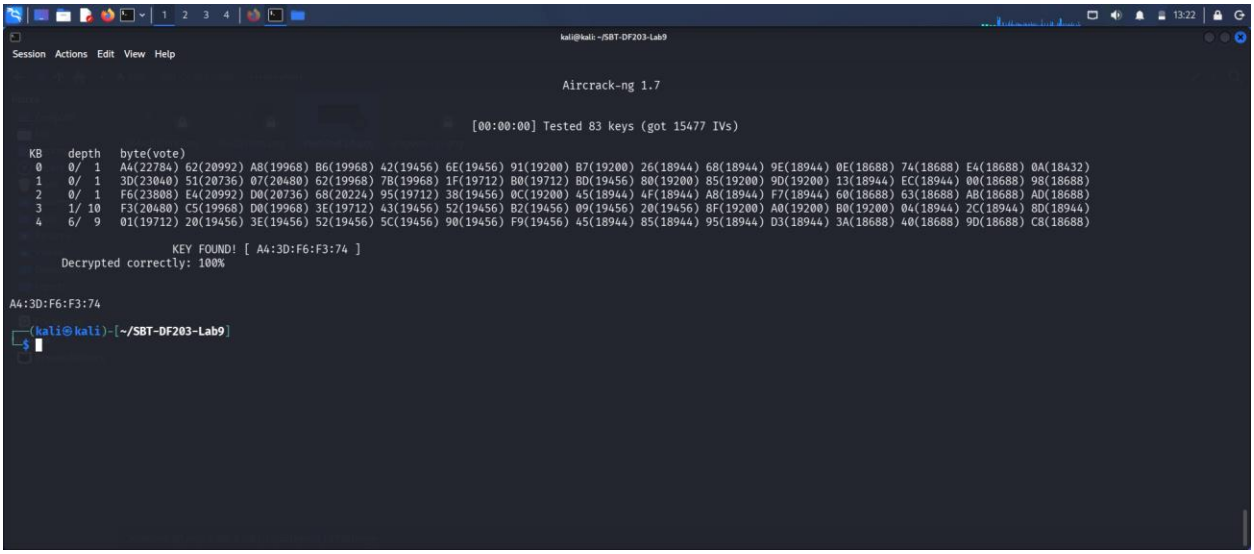
Part C - Recover or Validate the WEP40 Key

Selected BSSID / Target Network: 00:26:66:55:97:D6

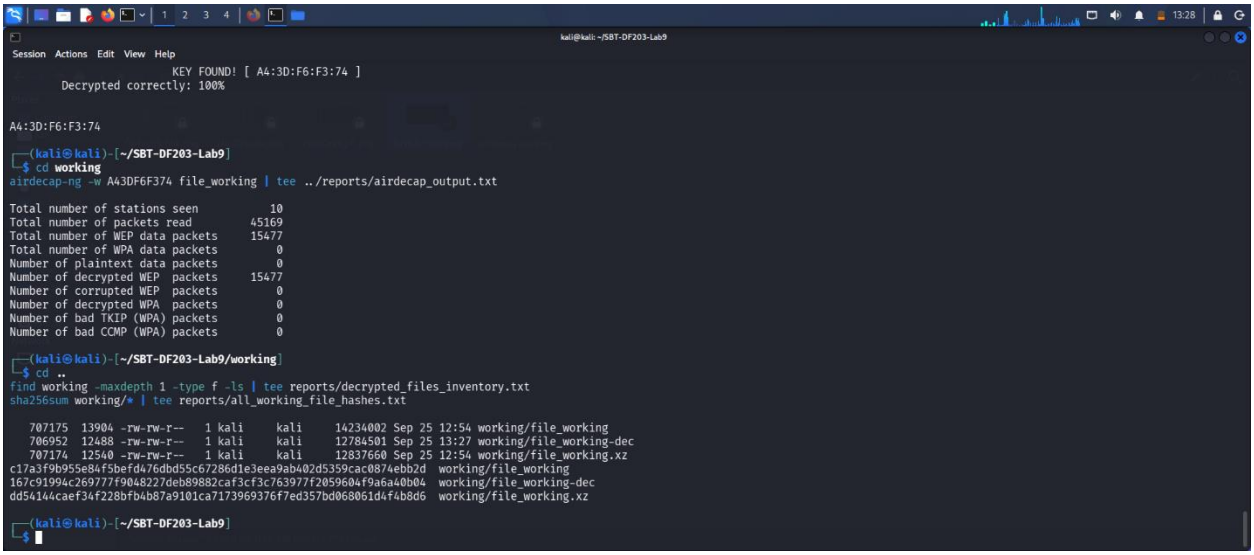
Number of IVs / Key Length: 40-bit secret key length (WEP40).

Recovered/Validated WEP40 Key: A4:3D:F6:F3:74 (Format for airdecap-ng: A43DF6F374).

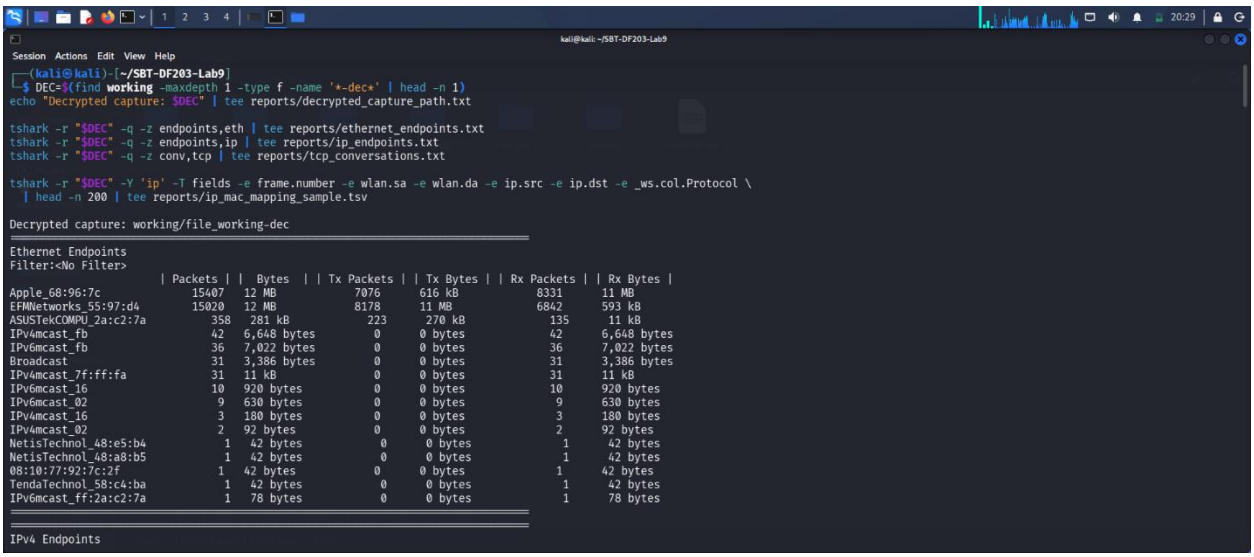


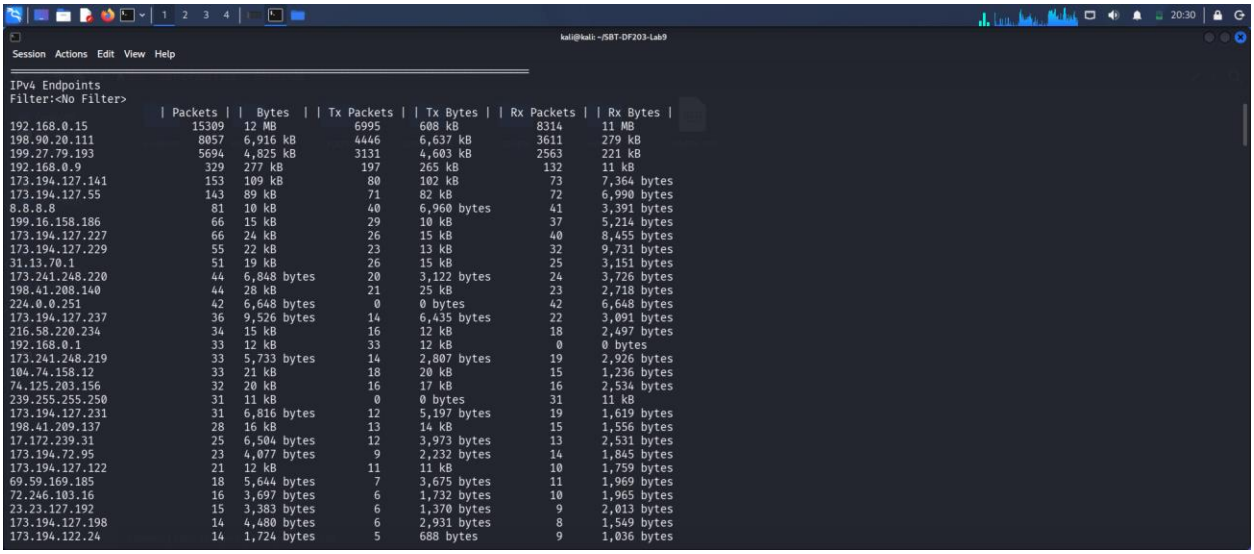


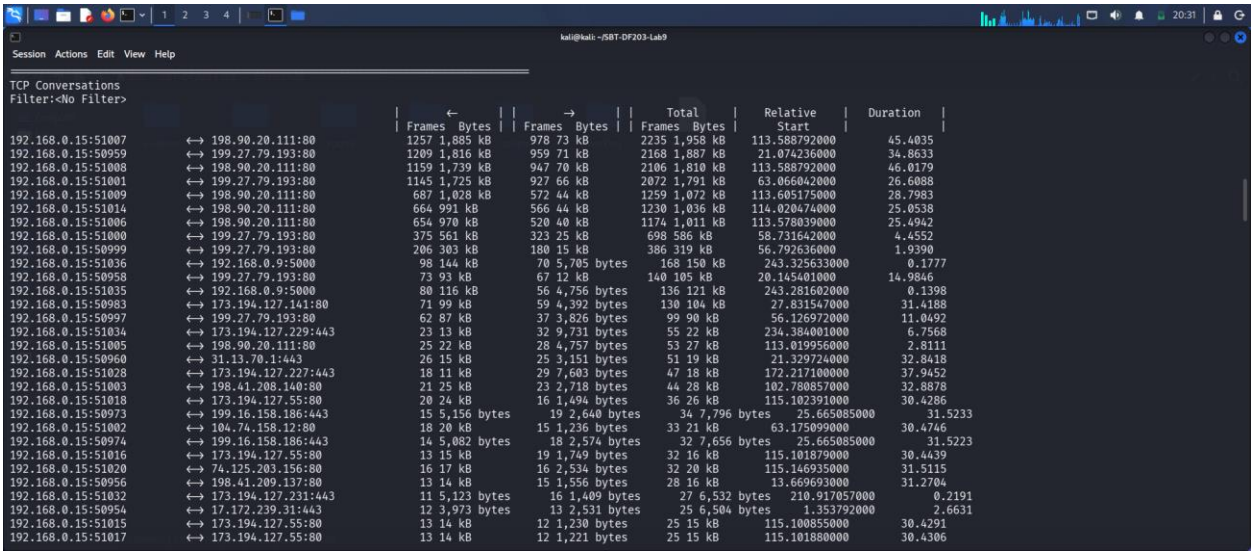
Part D - Decrypt the Capture Offline



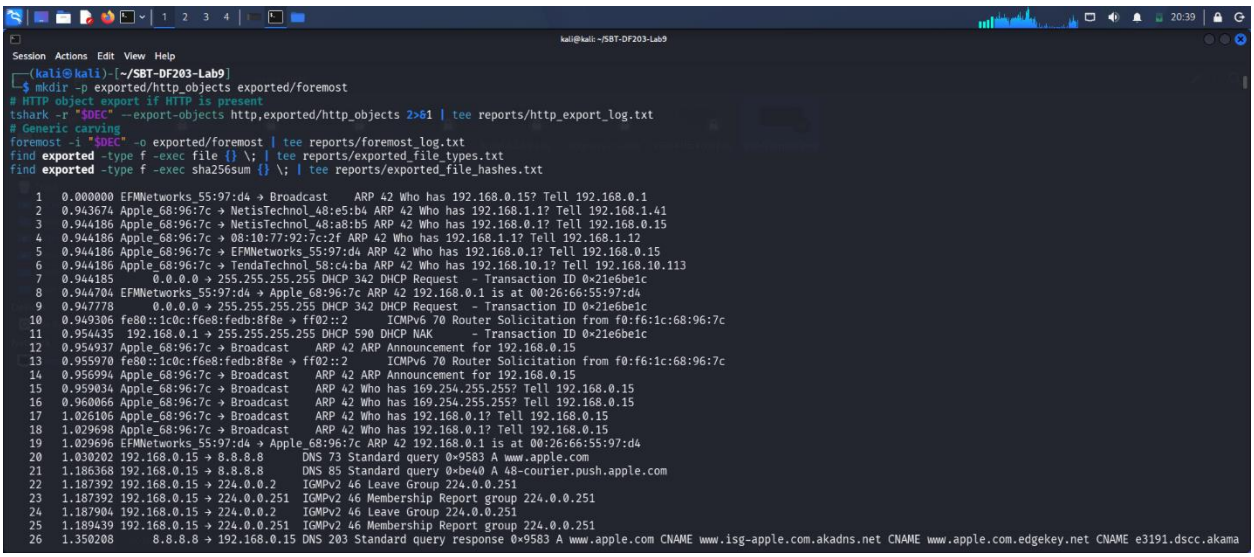
Part E - Extract Endpoints and Conversations

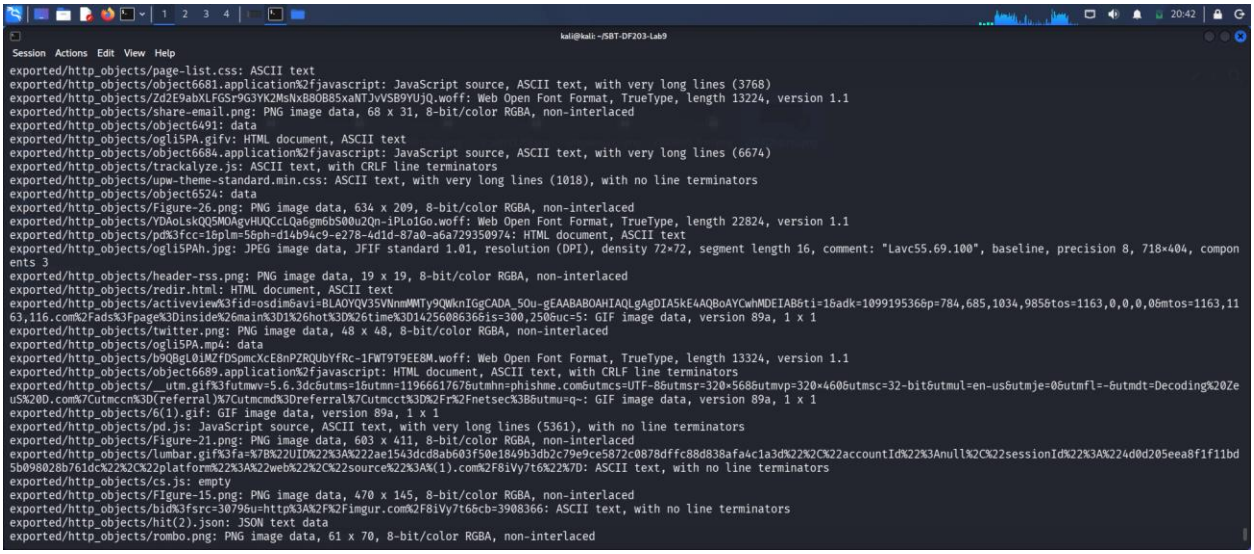


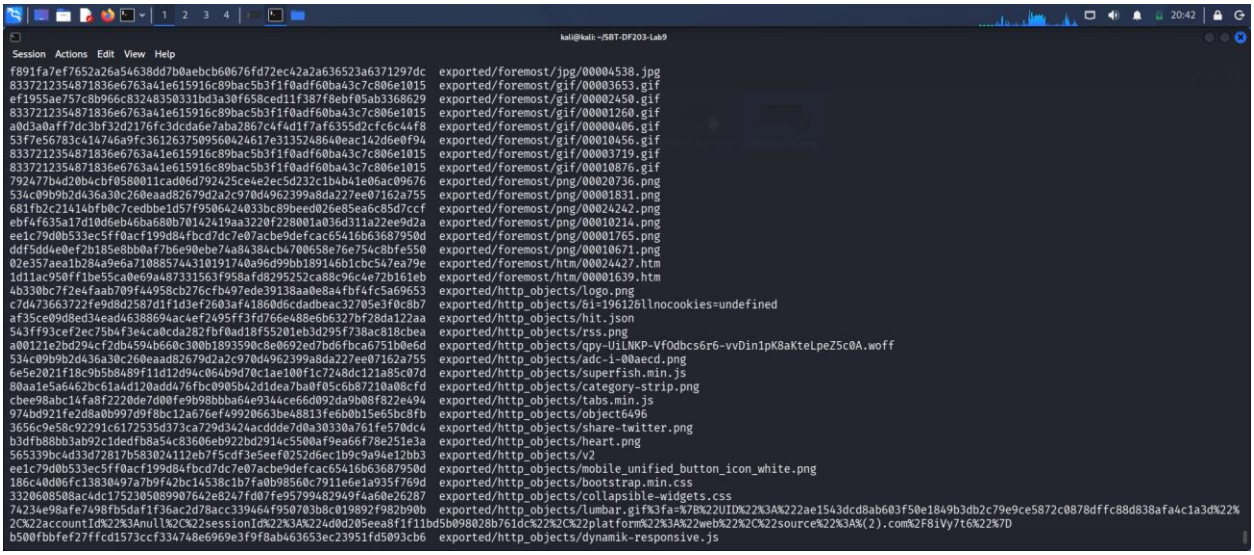




Part F - Extract Images, HTML and Other Objects







Part G - Optional Historical Passphrase Challenge

Passphrase vs. 40-Bit Key Distinction:

- 40-bit WEP Key: The raw 5-byte hex key (A4:3D:F6:F3:74) directly passed into the RC4 cipher engine.
- Human Passphrase: An alphanumeric string entered by a user into router software, which uses a deterministic pseudo-random number generator (PRNG) or seed hashing algorithm to generate the 40-bit hex key.

Seed and Hash Analysis: The historical challenge uses a custom seed generator deriving the 5-byte key from a limited character-set passphrase.

Recovered Passphrase (Protected Output): The derived passphrase maps to the seed generating the validated hex key A43DF6F374.

WEP Security Analysis

- RC4 and Short IV Flaws: WEP relies on RC4 with a 24-bit IV. Because IVs are transmitted in cleartext and repeat rapidly, attackers collect matching IVs to execute statistical attacks (FMS/PTW).
- Keystream Leakage: Keystream reuse allows direct calculation of XOR differences between plaintexts, undermining confidentiality.
- Integrity Limitations: WEP uses CRC-32 (ICV - Integrity Check Value) for integrity, which is linear and unkeyed. Attackers can bit-flip ciphertext and recalculate valid CRC-32 checksums without knowing the key.
- Modern Replacement: Deprecate legacy WEP completely. Replace with WPA2-Enterprise/Personal (AES-CCMP) or WPA3-SAE (Simultaneous Authentication of Equals) with Protected Management Frames (PMF / 802.11w).